

1-A:

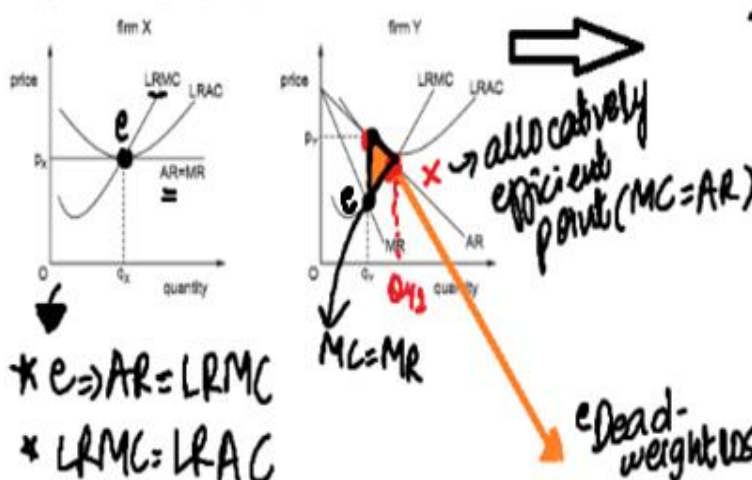
A – When a new firm is established in an area, and it **pays its workers higher wage rates than other firms**, this represents a “**private cost**” because prices or monetary values can be attached, as compared to externalities in the case of which, no prices are attached, because they do not go through the market. Since, it **does not** represent an externality, **this becomes correct.**

B – When flowers planted by a householder in his garden results in the third party (in this case, the neighbors) to benefit from the pleasure provided by the flowers, this represents the **existence of “external benefits”**

C – The immunization of children against small pox, when leads to the third party to benefit (in this case, other children, who are prevented from contracting the infection), this represents the **existence of “external benefits”**

D – When the installation of security cameras in the city center results in reduced theft in that particular city center, and on the expense of this, results in thieves attacking elsewhere, which results in an **increased theft elsewhere**, this represents “external cost of the action of installation of security cameras by the city center authorities”

2-C:

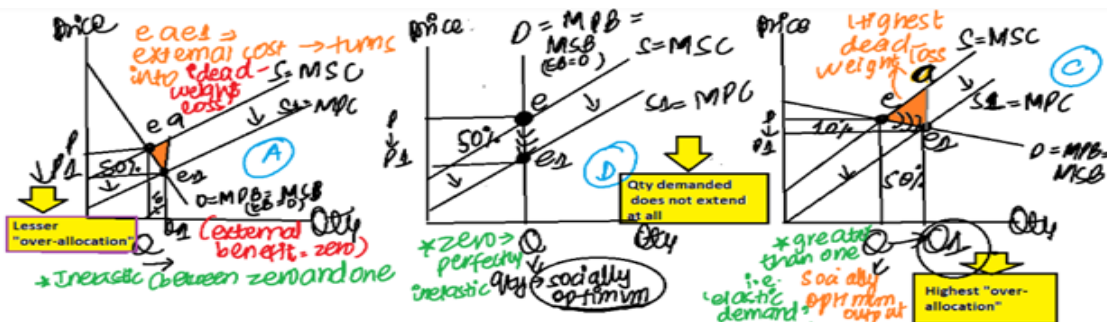
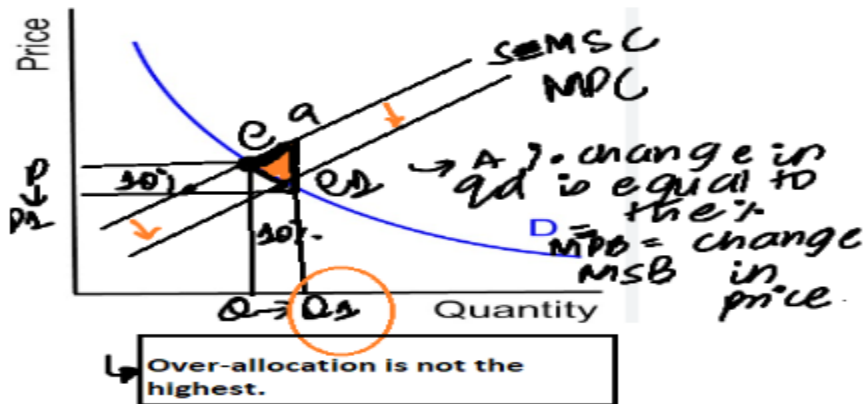


- When Average revenue (price) = Marginal cost (LRMC represents "long-run marginal cost"), this means that there is "allocative efficiency"
- When Marginal cost (LRMC) = Long Run Average cost (LRAC), this represents a situation whereby the marginal cost curve intersects the Average cost curve, on which, the Average cost is minimized. Here, productive efficiency is achieved.

- Firm Y is productively efficient because at Q_y level of output, cost cannot be further reduced, and in order to further decrease the cost, output will have to be increased. This means that it is "productively efficient"
- Allocative efficiency would occur at point "X" that is when LRMC (marginal cost) = AR (price). The firm's equilibrium is at point 'e', and it is currently producing Q_y level of output. The difference between the allocatively efficient point, and the point where $MC = MR$ represents the existence of "dead-weight loss of economic welfare". This means that the firm is not "allocatively efficient"

3-C:

Option B:



It can be seen in option C, that when prices are set low from the society's perspective, the quantity consumed is the highest when the demand is price-elastic, representing that consumption is too high from the society's perspective.

Note: The flatter the curve, the more elastic is the demand, and the higher will be the "dead-weight loss" that is created.

This is why, key C, is correct.

When highest over-allocation is done, the dead-weight loss will also be the highest.

4-B:

Allocative inefficiency occurs when the market fails to allocate resources efficiently because Marginal social costs are not equal to Marginal social benefits. This represents the existence of externalities that are costs or benefits that are not fully reflected in market prices.

Here's a brief explanation of the other options:

A. Allocative efficiency: This occurs when resources are allocated in a way that maximizes overall economic efficiency, that occurs when $MSB=MSC$.

C. Productive efficiency: This refers to the optimal production of goods and services given the available resources and technology.

D. Productive inefficiency: This occurs when firms produce goods and services at a higher cost than necessary, resulting in a loss of economic efficiency.

5-D:

In order to maximize utility, the utility-maximizing condition or the equi-marginal principle must be established. This is established when the marginal utility per dollar spent on good X is equal to the marginal utility per dollar spent on good Y. ($MU_x/P_x = MU_y/P_y$)

$$\begin{aligned} & \frac{\text{Marginal utility } X}{\text{Marginal utility } Y} = \frac{\text{Price of } X}{\text{Price of } Y} \\ & \frac{MU_x}{P_x} = \frac{MU_y}{P_y} \end{aligned}$$

6-B:

In this option, the budget line is shifting parallelly inwards, which represents that the **real incomes have fallen**.

And because of this, the quantities consumed of both: good X and good Y have decreased.

A – This is incorrect as the quantities of both the goods that are consumed are decreasing.

B – It is correct to say that the consumer prefers fewer of good X and fewer of good Y, because a fall in real incomes has led to a fall in the quantities demanded of both the goods.

C – No, this is incorrect to say that the consumer prefers **more of Y**, because the quantities that are demanded of both the goods fall.

D – This is incorrect as the quantities of both the goods that are consumed are decreasing.

7-A:

A – The key shows that “tick” represents all the points that are on the budget line, and here all the three combinations have a tick, which means that all the options are on the budget line, which makes it correct.

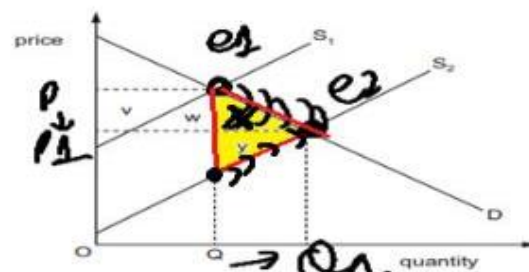
B – Here, there is one tick, and there are two crosses, which means that one point is on the budget line, whereas all others are not on the budget line.

C – Here, there are two ticks, and there is one cross. This means that two points are on the budget line, and one point is not on the budget line, which is why, this becomes incorrect.

D - Here, there are two ticks, and there is one cross. This means that two points are on the budget line, and one point is not on the budget line, which is why, this becomes incorrect.

8-C:

The diagram shows an industry's demand and supply curves. The industry increases supply from S_1 to S_2 .



Which area shows the total gain in society's economic welfare?

- A $v + w + x$ B $w + x$ C $x + y$ D $(x + y) - (v + w)$

This explains why, key C, will be correct.

9-A:

A: This is correct because this shows that **average revenue is greater than marginal revenue**, and importantly, **when marginal revenue is zero, that is when, total revenue is at its maximum.**

B: Here, the total revenue is a linear, upward sloping line, beginning from the origin, which represents the case of “perfect competition”. And, importantly, average revenue is shown to be lesser than marginal revenue here, which is incorrect since it needs to be greater than marginal revenue.

C: This is incorrect because this shows that , when Average revenue is zero, total revenue is maximized. Instead , total revenue is maximized when marginal revenue is zero.

D: Here, average revenue is shown to be lesser than marginal revenue , which is incorrect since it needs to be greater than marginal revenue. And, this also shows that when Average revenue is zero, total revenue is maximized. Instead , total revenue is maximized when marginal revenue is zero.

10-C:

Sales maximization is achieved when the highest number of units are sold. This occurs when the firm charges the lowest price it possibly can.

It charges the lowest possible price, when it makes "normal profits", that are made when **Average revenue=Average cost**.

Normal profits, also known as zero economic profits, refer to the level of profit earned by a firm that is just sufficient to cover all of its costs, including both explicit costs (such as wages, rent, and raw materials) and implicit costs (such as the opportunity cost of the owner's time and capital). In other words, normal profits represent the minimum level of profit required to keep a firm operating in the long run. Normal profits are considered the baseline level of profit that a firm needs to earn to remain in business.

Therefore, if the equilibrium qty of Q4 is determined through the equilibrium point where "AR=AC", we will obtain the equilibrium price of "OP4" through the Average revenue curve.

At the price of OP4, the highest units of Q4 are being sold, this is why, key C is correct.

11-C:

C - The data shows that in both 2015 and 2016, the number and value of purchases by UK companies (both abroad and in the UK) exceeded the number and value of sales by UK companies. This indicates that UK companies were net buyers in both years, meaning they acquired more assets or companies than they sold.

Here's a brief explanation of why the other options are incorrect:

A. Purchasing activity abroad by UK companies actually decreased from 2015 to 2016, from 170 transactions worth £25,561m to 135 transactions worth £16,665m.

B. While UK companies made more purchases than foreign companies in 2015, the opposite was true in 2016, when foreign companies made more purchases than UK companies.

D. The value of sales by foreign companies actually decreased from 2015 to 2016, from 7451 million pounds to 4807 million pounds.

12-D:

We are interested in an option in the case of which, the size of the industry is shown to rise, and there is a fall in the average output or the average product.

A – An advancement in communication methods which reduces communication problems within the firm will lead to an increase in efficiency, and in turn, a rise in the Average product.

B_ – An advancement in technology which increases the cost advantages of a firm is representative of a rise in productivity and in turn, a rise in the Average product.

C- If there is an increase in the barriers to entry and exit, the size of the industry will be small.

D-An increase in external economies of scale represents that the ^{industry} has grown large in size. And, a decrease in internal economies of scale explains that the average product or the average output has fallen, which is why, now internal economies of scale are not being exploited.

13-B:

A - A large number of firms in an industry makes collusion harder, not easier. With many players, it's more difficult to coordinate and maintain agreements, as each firm may have different interests and incentives.

B. Homogenous products mean that firms are producing identical goods. When firms produce homogenous products, it makes it easier for them to collude. This is because this makes it easier for firms to agree on prices and output levels, as they have a common interest in maintaining market share.

C. Low barriers to entry make collusion harder, not easier. When it's easy for new firms to enter the market, it increases competition and makes it harder for incumbent firms to maintain agreements. New entrants may disrupt the market as they can increase the market output because of which, market prices can fall.

D. Rapid technological change can disrupt collusion, not facilitate it. When technology is constantly evolving, firms need to adapt quickly to stay competitive. This can make it hard to maintain agreements, as firms may need to adjust their strategies to keep up with changing market conditions.

14-B:

Transfer Earnings:-

These are the earnings the fashion model could get in his next best alternative job, which is teaching. In this case, his transfer earnings are \$100,000 per year.

Economic Rent:-

This is the excess amount the fashion model earns above his transfer earnings, due to his unique skills, abilities, or circumstances that make him particularly valuable in his current job. In this case, his economic rent is the difference between his current earnings and his transfer earnings:

$$\begin{aligned}\text{Economic Rent} &= \text{Current Earnings} - \text{Transfer Earnings} \\ &= \$500,000 - \$100,000 \\ &= \$400,000 \text{ per year}\end{aligned}$$

So, the fashion model's transfer earnings are \$100,000 per year, and his economic rent is \$400,000 per year.

15-D:

➔ If wages in industry X are significantly higher than in industry Y, it could be because industry Y offers non-monetary benefits that compensate for the lower wages, such as: **better working conditions, job security and flexible working hours**. This means that while workers in industry Y may earn lower wages, they receive other benefits that make up for the difference, thereby making option D correct.

A. If workers in industry Y are highly mobile, they would likely move to industry X for better pay, which would reduce the wage difference.

B. Strong trade union organization in industry Y would likely lead to higher wages, not lower.

C. If industries X and Y compete for workers, wages in industry Y would likely increase to match those in industry X.

16-C:

C) The government taxes producers of good X and PED of demand for good X is elastic:

-> If the government taxes producers of good X (which is a “demerit good”, which generates negative externalities), then after prices are raised because of indirect taxes (because of a decrease in supply, which creates a shortage, and because of a producer’s attempts to pass burden to consumers in form of higher prices), quantity demanded will contract by a relatively higher percentage if the demand for them is elastic(>1)

-> And when the demand for the other good i.e. good Y (the substitute with fewer negative externalities) is also elastic (>1) , a greater proportion of consumers will switch to good Y which generates lesser external costs.

-> Importantly, subsidies should not be given to producers of good Y because the government is not wishing for, or aiming for a “budget deficit” This is why, key C, is correct.

17-B:

Direct taxes will fall:

When direct taxes (that are progressive in nature as in their case the average rate of tax rises upon a rise in income and the marginal rate of tax is greater than the average rate of tax which represents that they put up a greater burden on the rich) are reduced, the government will have less tax revenues for redistribution to the lower income groups in form of means tested benefits.

As a result of this, inequalities will rise.

Indirect taxes rise:

When indirect taxes (that are regressive in nature as they take up a greater proportion of the incomes of the poor than the rich because the average rate of tax falls upon a rise in income) are raised, it will increase the prices of essential food items, and since their demand is inelastic, qty demanded by the low income groups will contract by a relatively smaller percentage, ultimately leading to a rise in the expenditures of the low income groups.

This also contributes to an increase in inequality.

18-B:

→ Nudge theory, also known as behavioral economics, involves using indirect and non-coercive methods to influence people's behavior and decision-making. The goal is to guide individuals towards making choices that are in their own best interests or align with broader societal objectives. Nudge theory emphasizes on the importance of actual behavior in an economy, according to which, the economic behavior of individuals can be “nudged” in a particular direction.

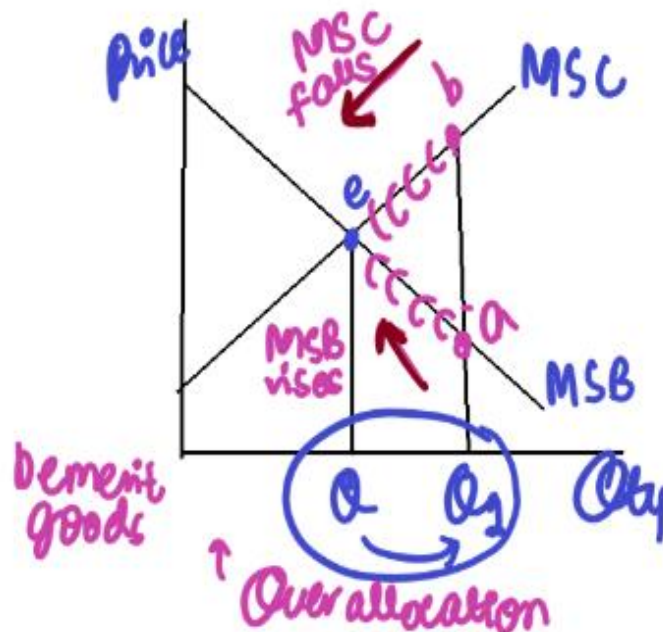
B – When signs are installed that direct people to new litter bins, individuals will be persuaded or poked to change their direction towards the litter bin to throw the litter, without being imposed. And, a gesture of gratitude for usage of the bins will also prove to psychologically influence them. This is why, key B, involves the usage of “nudge theory”

A – Banning the dropping of litter represents a “regulation”, which is not a “nudge”, as you don’t impose or restrict under nudge.

C – Raising local tax and using the money to employ more people to pick up litter again does not involve a nudge.

D – Neither is “removing all litter bins” a part of nudge, nor will it help to improve the situation of “littering”, as it will instead prove to worsen it.

19-D:



A- This is incorrect because MSB rises when quantity is contracted.

B- Again, Marginal social benefit rises when quantity is contracted.

C - If the fall in the total social benefit is greater than the total social cost, this is a disadvantage, which will result in a fall in "society's net welfare"

D - If the fall in total social cost is greater than the fall in the total social benefit, this represents a "fall in external cost", as a result of which, society's net social welfare increases.

20-D:

- ➔ Real GDP per capita is a measure of the inflation-adjusted output of a country divided by its population. However, it does not capture income inequality within the country. If a small portion of the population earns a significant share of the total income, the GDP per capita figure may not accurately represent the living standards of the majority of the population.

In the case of New Zealand, a GDP per capita of \$34,762 may not reflect the actual living standards of many individuals, especially if there are significant disparities in income distribution.

This explains why key D will be correct.

A. Foreign trade(X-M) is actually included in the GDP calculation, as it affects the country's overall production and income.

B. Changes in population are already accounted for in the per capita calculation, which divides the total real GDP by the total population.

C. The question mentions "real" GDP, which means that inflation has already been taken into account. Real GDP is adjusted for price changes to reflect the actual increase in goods and services produced.

21-C :

Sustainable economic growth means that the growth is not only stable and long-term but also considers the well-being of future generations. It balances economic progress with social and environmental considerations.

C - In this context, achieving sustainable economic growth would mean that the growth is not only beneficial for the current population but also ensures a prosperous future for generations to come.

A. Sustainable economic growth does not necessarily mean that all countries would benefit equally from global economic growth.

B. It does not imply that all employees would receive the same growth in incomes. Equality is a different concept than sustainable economic growth.

D. It does not guarantee that all producers would experience the same growth in demand.

22-D:

A - The dependency ratio refers to the ratio of dependents (children and elderly) to working-age individuals. There's no information in the table to suggest a change in the dependency ratio.

B. There's no information about job vacancies in the table, so we can't conclude anything about them.

C - The participation rate refers to the proportion of the population that's actively engaged in the workforce. We are not aware of the population size to account for this.

D - The table shows that the number of employed individuals increased from 10 million in 2012 to 11 million in 2017, and also, the number of unemployed also rose denoted through a rise in the unemployment rate. This represents a rise in labor force, which indicates a rise in the working population. This is why, this becomes correct.

23-D:

When workers are unemployed due to technological changes, this represents technological unemployment which is a part of structural unemployment. This type of unemployment is a result of structural changes in the economy, making it difficult for workers to adapt and find new employment. This is why key D is correct.

A. Cyclical unemployment refers to unemployment caused by economic downturns or recessions, not technological change.

B. Frictional unemployment is a temporary phenomenon that occurs when workers are transitioning between jobs.

C - Seasonal unemployment is a type of unemployment that occurs due to seasonal fluctuations in industries such as agriculture, tourism, or retail.

24-A:

$$\begin{aligned}
 & \star \text{ Change in real income/head} \Rightarrow \text{Change in national income} - \text{Change in prices} - \text{Change in population size} \\
 & \star \Rightarrow +5\% - 4\% - 2\% \\
 & \Rightarrow \textcircled{-1\%} \rightarrow \text{Decrease of } 1\%
 \end{aligned}$$

25-A:

Commercial banks in a country hold \$50 million in cash but have created \$600 million deposits.
The central bank raises the cash deposit ratio to 10%.

What will be the outcome?

$$\star \text{ Initial cash deposit ratio} = \frac{50}{600} \times 100 \Rightarrow 8.33\%$$

$$\star \text{ New cash deposit ratio} = \frac{n}{600} \times 100 \Rightarrow 10$$

$$\hookrightarrow \frac{100n}{600} = 10\%$$

$$\hookrightarrow \frac{1}{6}n = 10 \div \frac{1}{6}$$

$$n = \textcircled{60}$$

total
When deposits are unchanged, cash held will need to be increased to \$60, so that the cash deposit ratio is 10%

$$\begin{aligned}
 & \text{Cash deposit ratio} \Rightarrow \frac{\text{Cash} \times 100}{\text{Deposits}} \\
 & \textcircled{\text{Cash deposit ratio increases when}} \\
 & 1) \text{ cash } \uparrow \\
 & 2) \text{ deposits } \downarrow
 \end{aligned}$$

Commercial banks in a country hold \$50 million in cash but have created \$600 million deposits.
The central bank raises the cash deposit ratio to 10%.

What will be the outcome?

* Initial cash deposit ratio = $\frac{50}{600} \times 100 \Rightarrow 8.33\%$

* New cash deposit ratio = $\frac{50}{n} \times 100 \Rightarrow 10$

Let deposits be n
 $\hookrightarrow 5000 = 10n$
 $\hookrightarrow \frac{5000}{10} = n$
 $n = 500$

When cash held is kept unchanged, deposits will have to be decreased to \$500 so that the new cash deposit ratio is 10%.
 [option A]

* Cash deposit ratio $\Rightarrow \frac{\uparrow \text{Cash} \times 100}{\downarrow \text{Deposits}}$
 @ cash deposit ratio increases when:
 1) cash $\uparrow$
 2) Deposits $\downarrow$

This explains why option A will be correct.

26-D:

Real income increases:

-> When real income rises, people have more money to purchase goods and services, as a result of which, the demand of money for transactionary purposes will increase.

Price level increases:

-> When price level increases, the real value of money decreases, which means that more money will be required to purchase the same amount of goods and services. This will again result in an increase in the transactionary demand of money.

Interest rates decrease:

-> When interest rates decrease, savings fall because of a decrease in the return on savings. And so, individuals will be encouraged to consume more, thereby bringing about a rise in the quantity of money which is demanded.

27-B:

If the tariffs on imports from Zambia are removed, then since Zambian exports appear more internationally competitive, as their prices are now lowered, there will be an extension in the international qty demanded of copper by a relatively higher percentage when its demand is elastic ($PED > 1$), which will result in a rise in export revenues of Zambia, thus leading to them being benefitted more. This is why, B is correct.

Income-elasticity of demand is not a correct option because a reduction in tariffs will affect prices of copper, rather than incomes.

* Tariffs $\downarrow$ (10%)
 * price of zambian copper $\downarrow$ (10%)
 * Qty of copper $\uparrow$ (50%) $\rightarrow$ $ped > 1$ (elastic demand)
 * X (TR) $\uparrow \rightarrow$ zambia

28-C:

After appreciation, since the prices of export-oriented goods rise, and they become uncompetitive, their quantity demanded will contract and that too by a relatively higher percentage when their demand is elastic, which will result in a fall in export-revenues. This fall in export revenues represent the fact that the Balance of payments on the current account position **“worsens”**

Additionally, when import prices fall, but the demand for imports is inelastic (since we are importing raw materials), the qty demanded for imports will extend by a relatively smaller percentage, resulting in a fall in import expenditures. This means that when costs of production fall, and subsequently Aggregate supply rises, this results in **the rate of “cost-push inflation”** to fall.

This is why, key C, becomes correct.

29-D:

A) This option is incorrect because when workers are unemployed, their skills and experience can become outdated, leading to a decline in their human capital. Prolonged unemployment can also lead to a loss of confidence, networking opportunities, and access to training and development, further eroding their human capital. **This explains the “hysteresis effect”**

B) This option is incorrect because rising unemployment typically leads to individuals not being able to access a wide variety of high quality consumer goods. Additionally, because of a loss of incomes, they may not be able to access healthcare services, and also, unemployment contributes to an increase in social unrest. This explains how living standards will fall in event of rising unemployment.

C) The surplus of available workers in event of unemployment will exert a downward pressure on nominal wages.

D) This option is correct because when unemployment rises, the government's tax base shrinks, leading to lower tax revenues. To maintain public services and support for the unemployed, the government may increase tax rates or impose new taxes on employed workers, thereby increasing their tax burden.

30-D:

30 The table gives some economic indicators for four countries.

Which country is closest to achieving the principal aims of government economic policy?

country	change in GDP (%)	rate of inflation (%)	unemployment (%)	balance of payments (% of GDP)
A	0.47	0.8	3.4	+3.3
B	1.13	0.9	10.5	-0.1
C	1.18	1.1	6.9	-3.3
D	1.45	0.1	4.6	+8.5



- > The %increase in GDP is the highest of about 1.45% (D)
- > The rate of inflation is the lowest of 0.1%
- > Unemployment % is the second lowest of about 4.6%
- > The Balance of Payments surplus (as a % of GDP) increased by the highest percentage(8.5%)

Macroeconomic Objectives → economic growth
→ price stability
→ Full employment
→ Balance of payments surplus
→ Redistribution of income and wealth

